

CS 232 Operating Systems

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(Tentative) Syllabus is online.

Class etiquette:

- Be punctual
- Respect others
- Try to participate in discussions

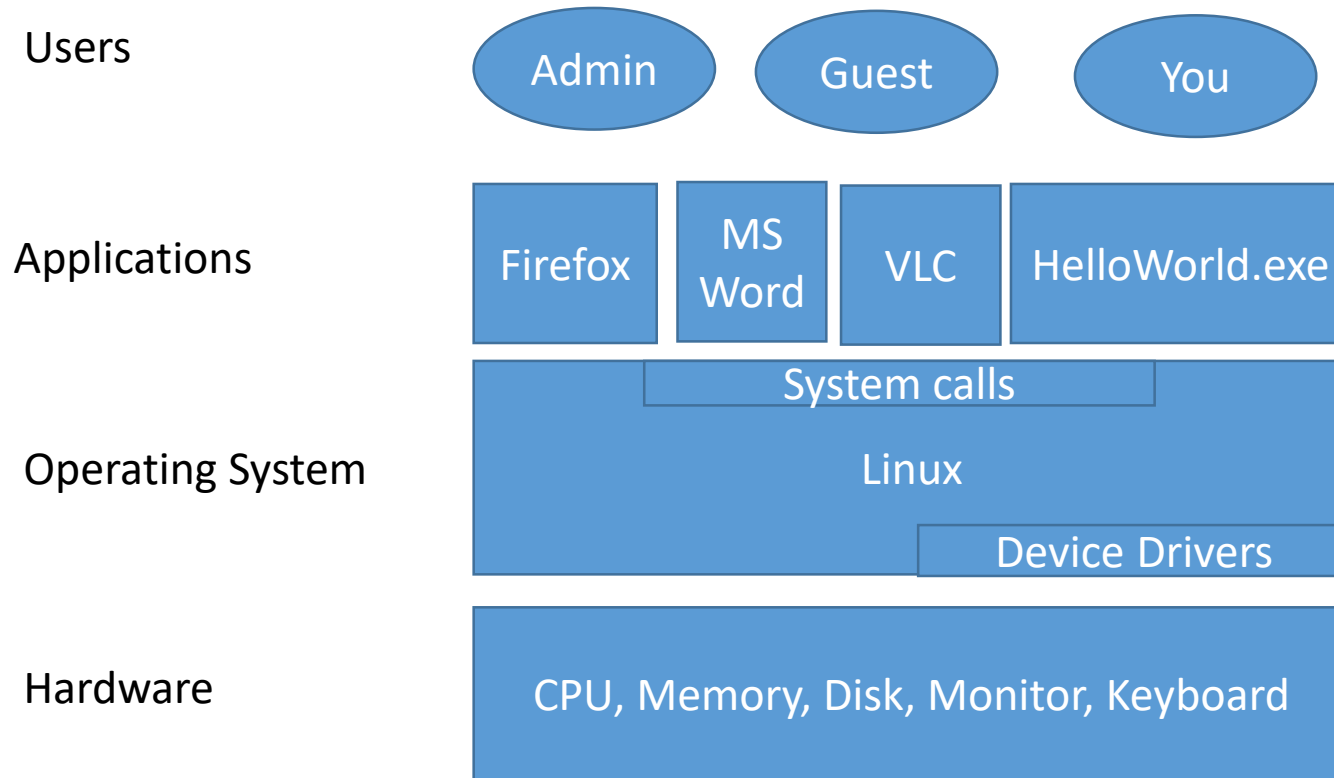
Know any OS names?

There are tons!



What is an Operating System

The scheme of things

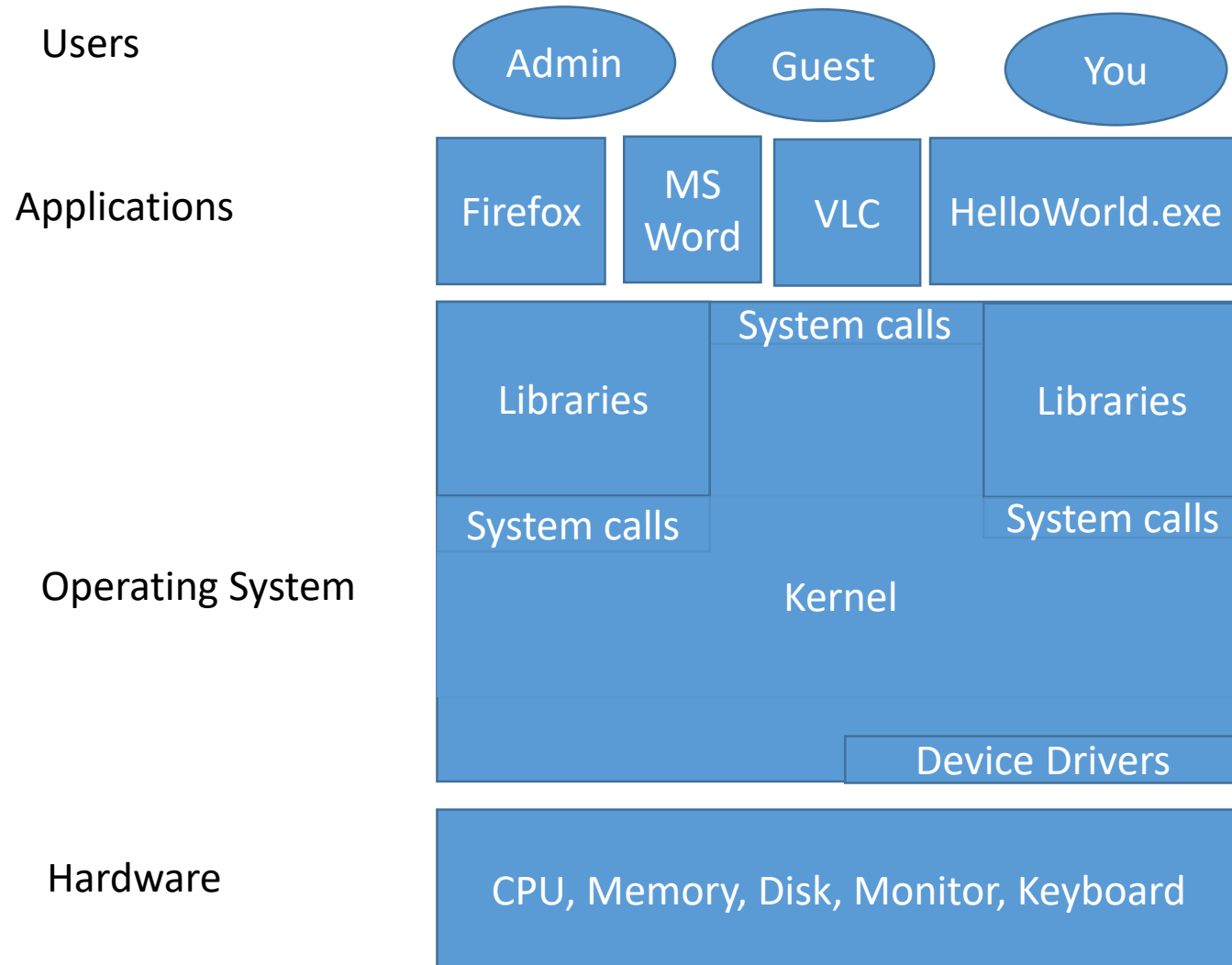


What happens
when you:

Execute:
`printf("Hello World!\n");`

Double click on a film?

The scheme of things



Linux distribution =

Kernel + Libraries +
Applications

What does an OS do?

Basically a Resource Manager. It manages:

- Hardware
 - . CPU
 - . Memory
 - . Storage
 - . I/O
- Mechanisms for communication b/w running programs
- Protection and Security (among Users/Programs and from outside)

Design goals

- Convenience
- Performance
- Protection
- Security
- Energy Efficiency

A history

- C. 1950s-60s: Computers are custom made. Come with own software. Programming is in Assembly. OS were libraries. Operators for Computers. Single Program runs. Batch Processing. MULTICS.
- C. 1970s: Computers start sharing ISAs. Minicomputers. IBM System/360; OS/360. Ken Thompson develops Unix for PDP-7, PDP-11. C Language.
- C. 1980s: PCs. Bell Labs “sells” Unix. Stallman; GNU/Hurd; FSF; GPL. BSD. MINIX. MS DOS.
- C. 1990s: Linux. MS Windows.
- C. 2000s: Mobile OS
- C. 2010s: Android, iOS, Others.
- Future : Cloud OS? Rust?